1. identify the stages in a given chemical synthesis of an inorganic compound including:
   - choosing the reaction or series of reactions to make the required product;
   - carrying out a risk assessment;
   - **working out the quantities of reactants to use;**
   - carrying out the reaction in suitable apparatus in the right conditions (such as temperature, concentration or the presence of a catalyst);
   - separating the product from the reaction mixture;
   - purifying the product;
   - measuring the yield and checking the purity of the product;
2. understand the purpose of these techniques: dissolving, crystallisation, filtration, evaporation, drying in an oven or dessicator;
3. understand the importance of purifying chemicals and checking their purity;
4. understand that a balanced equation for a chemical reaction shows the relative numbers of atoms and molecules of reactants and products taking part in the reaction;
5. understand that the relative atomic mass of an element shows the mass of its atom relative to the mass of other atoms;
6. be able to use the Periodic Table to obtain the relative atomic masses of elements;
7. calculate the relative formula mass of a compound using the formula and the relative atomic masses of the atoms it contains;
8. **calculate the masses of reactants and products from balanced equations;**
9. calculate percentage yields given the actual and the theoretical yield;
10. describe how to carry out an acid-alkali titration accurately (solid sample weighed out into a titration flask, dissolved in water and then titrated with acid or alkali from a burette);

ⓘ Making up of standard solutions is not required.

11. substitute results in a given formula to interpret titration results quantitatively.

12. understand why it is important to control the rate of a chemical synthesis (to include safety and economic factors);
13. explain the term: ‘rate of chemical reaction’;
14. describe methods for following the rate of a reaction (for example by collecting a gas, weighing the reaction mixture or observing the formation of a colour or precipitate);
15. interpret results from experiments that investigate rates of reactions;
16. recall that reaction rates vary with the particle size of insoluble chemicals, the concentration of solutions of soluble chemicals and the temperature of the reaction mixture;
17. understand that catalysts speed up chemical reactions while not being used up in the chemical changes;
18. interpret information about the control of rates of reaction in chemical synthesis;
19. **use simple collision theory to explain how rates of reaction depend on the concentration of solutions of soluble chemicals.**

The five topics in this longer module introduce new chemical ideas while illustrating important features of the applications of chemistry and exploring Ideas about Science from IaS1 Data and its limitations, IaS3 Developing explanations, IaS5 Risk, and IaS6 Making decisions about science and technology.

The first topic covers introductory organic chemistry taking alcohols and carboxylic acids as the main examples. This builds on the coverage of hydrocarbon molecules in modules C1 Air pollution and C2 Material choices. The second and third topics lay the foundations for more advanced stud of physical chemistry by exploring chemical concepts on a molecular scale include the connection between energy changes and bond breaking as well as the notion of dynamic equilibrium.

The fourth topic introduces concepts of valid analytical measurements in contexts where the results of analysis matter. The two main analytical methods featured are chromatography and volumetric analysis. The final topic covers green chemistry and describes how the chemical industry is reinventing processes so that the manufacture of bulk and fine chemicals is more sustainable.

Topics	
C7.1	Alcohols, carboxylic acids and esters
Organic molecules and functional groups; alcohols; carboxylic acids; esters.	
C7.2	Energy changes in Chemistry
Why are there energy changes during chemical reactions?	
C7.3	Reversible reactions and equilibria
Introducing dynamic equilibrium	
C7.4	Analysis
Analytical procedures; chromatography; quantative analysis	
C7.5	Green Chemistry
The chemical industry; the characteristics of green Chemistry; making ethanol.	

ICT Opportunities

This module offers opportunities for illustrating the use of ICT in science, for example:

- modelling the structures of molecules;
- the integral role of ICT in chemical instrumentation.

Use of ICT in teaching and learning can include:

- modelling software to explore the shapes of organic molecules;
- video clips to illustrate the manufacture of chemicals on large and small scales;